Coffee Shop Network

Cisco Packet Tracer - Step-by-Step Build Guide (Project 1)

Goal: build a secure office for the business, a reliable system to process customer payments, and free Wi-Fi for customers.

You will use: 1 Router, 1 Multilayer Switch, 1 Access Point, 1 PC, 2 Printers, 1 POS terminal.

1. What you are building

Area	Devices	VLAN
Manager Office	Manager PC + Office Printer	10
Checkout Counter	POS Terminal + Receipt Printer	20
Guest Wi-Fi	Wireless Access Point	30
Small Network Cabinet	Router + Multilayer Switch + ISP connection (Fiber / Coaxial)	99 (management)

Note: A POS (Point of Sale) terminal is the device or system a business uses to accept payments and complete sales.

ISP connection methods

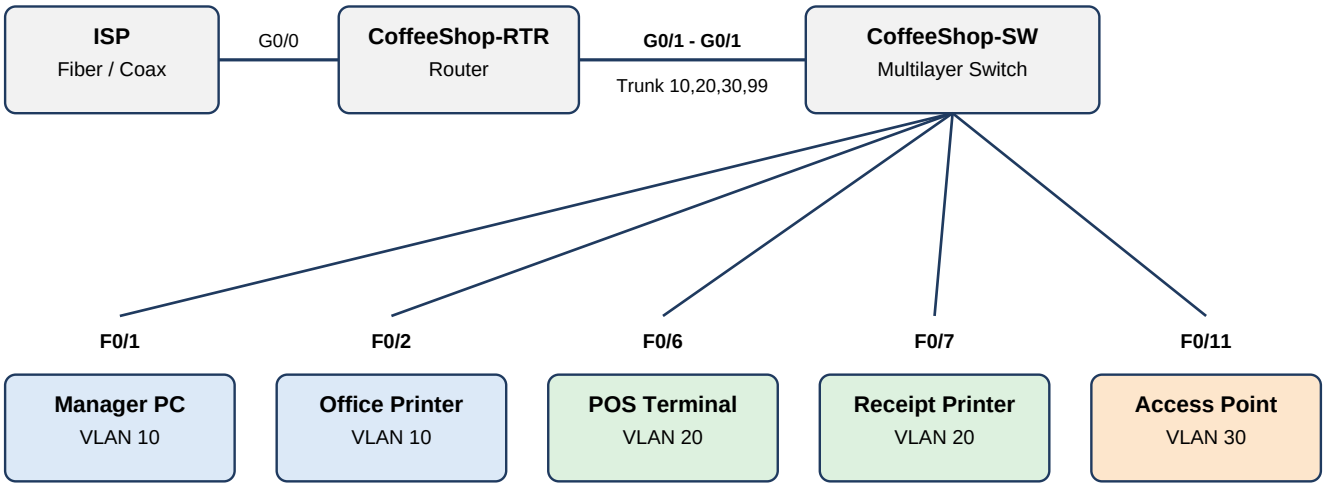
Method	Path	When
1) Fiber + ONT	ISP Fiber - Termination Box - ONT - Ethernet - Cisco Router	Common with modern fiber connections
2) Direct Fiber	ISP Fiber - Cisco Router SFP port	When the router and the ISP support direct fiber

2. Network plan (cheat sheet)

VLAN	Name	Subnet	Gateway (router)	Switch ports	Addressing
10	Management-Office	192.168.10.0/24	192.168.10.1	F0/1 - F0/5	DHCP (.21+); printers static
20	POS	192.168.20.0/24	192.168.20.1	F0/6 - F0/10	DHCP (.21+); printer static
30	Guest-WIFI	192.168.30.0/24	192.168.30.1	F0/11	DHCP (.21+)
99	Network-Management	192.168.99.0/24	192.168.99.1	none (switch SVI)	Static: switch = 192.168.99.2

First 20 addresses of each DHCP subnet (.1 - .20) are reserved for gateways, printers and future static devices.

Topology



3. Physical connections

#	From	To	Cable
1	Router G0/0	ISP	Ethernet
2	Multilayer Switch G0/1	Router G0/1	Ethernet
3	Manager PC	Switch F0/1	Ethernet
4	Office Printer	Switch F0/2	Ethernet
5	POS Terminal	Switch F0/6	Ethernet
6	Receipt Printer	Switch F0/7	Ethernet
7	Access Point	Switch F0/11	Ethernet

Note: Real life: the PC connects to a wall jack, and cables run across the ceiling to the switch. The Access Point gets power and network over one Ethernet cable using PoE (Power over Ethernet).

4. Switch configuration (CoffeeShop-SW)

Steps 1-5: Basic setup and security

Hostname, no DNS lookup, secret password, password encryption, day banner, console password.

```
enable
configure terminal
hostname CoffeeShop-SW
no ip domain-lookup
enable secret YOUR_SECRET
service password-encryption
banner motd #Authorized access only#
line console 0
 password YOUR_CONSOLE_PASS
 login
 exit
```

Step 6: Create the VLANs

Separate the office, POS, guest Wi-Fi and management traffic.

```
vlan 10
 name Management-Office
vlan 20
 name POS
vlan 30
 name Guest-WIFI
vlan 99
 name Network-Management
 exit
```

Step 7: Assign ports to VLANs + PortFast

Interfaces 1-5 = VLAN 10, 6-10 = VLAN 20, 11 = VLAN 30. PortFast makes end-device ports come up fast.

```
interface range f0/1 - 5
 description ## Management-Office Devices ##
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 exit
interface range f0/6 - 10
 description ## POS Devices ##
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 exit
interface f0/11
 description ## Guest Wireless Access-point ##
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 exit
```

Step 8: Trunk to the router

One physical link carries all VLANs using VLAN tagging. Allowed list limits the trunk to VLANs 10, 20, 30, 99 (security).

```
interface g0/1
 description To Coffeeshop RTR
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,99
 exit
```

Note: If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line.

Steps 9-10: Management SVI and default gateway

The switch gets its own management address in VLAN 99.

```
interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 192.168.99.1
```

Step 11: SSH for secure remote management

SSH needs a hostname, a domain name, an RSA key and a local user. The domain name below is just an example.

```
ip domain-name coffeeshop.local
username admin privilege 15 secret [secret]
crypto key generate rsa modulus 1024
ip ssh version 2
line vty 0 15
 transport input ssh
 login local
 exit
```

Save the configuration

So it survives a power loss.

```
end
copy running-config startup-config
```

Note: Best practice: leave spare interfaces for each department, and keep network management on its own VLAN.

5. Router configuration (CoffeeShop-RTR)

Steps 1-5: Basic setup and security

Same pattern as the switch.

```
enable
configure terminal
hostname CoffeeShop-RTR
no ip domain-lookup
enable secret YOUR_SECRET
service password-encryption
banner motd #Authorized access only#
line console 0
 password YOUR_CONSOLE_PASS
 login
 exit
```

Steps 6-7: Router interfaces (ISP side and switch side)

G0/0 faces the ISP. G0/1 faces the switch and carries the sub-interfaces (no IP on the physical port).

```
interface g0/0
 description ## To ISP ##
 ip address dhcp
 no shutdown
 exit
interface g0/1
 description ## To CoffeShop-SW ##
 no shutdown
 exit
```

Note: The notes do not give the ISP-side address. 'ip address dhcp' is an assumption - use the address your ISP/cloud gives you. The notes also do not include NAT; add it only if your lab needs it for Internet access.

Step 8: Router-on-a-stick (sub-interfaces)

One sub-interface per VLAN. Each one is the gateway for its VLAN.

```
interface g0/1.10
 description Management Office Gateway
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface g0/1.20
 description POS Gateway
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface g0/1.30
 description Guest-WIFI Gateway
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
interface g0/1.99
 description Network Management
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
 exit
```

Step 9: DHCP

The router hands out IP addresses automatically. The first 20 addresses of each subnet are reserved (infrastructure devices, printers, gateways, future static devices).

```
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20

! 1st pool - Management Office
ip dhcp pool MANAGEMENT_OFFICE
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8

! 2nd pool - POS
ip dhcp pool POS
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8

! 3rd pool - Guest Wi-Fi
ip dhcp pool Guest_WIFI
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
 exit
```

Review the pools with: `do show ip dhcp pool`

> **Note:** Big companies use a dedicated Linux/Windows DHCP server; the router forwards requests to it using DHCP Relay. Static IPs for infrastructure devices make management much easier.

Step 10: Guest ACL (Internet only)

The router can route between all VLANs, so block Guest Wi-Fi from reaching the Management, POS and Network-Management networks.

```
ip access-list extended GUEST_RESTRICTIONS
 permit udp any eq bootpc any eq bootps
 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
 permit ip 192.168.30.0 0.0.0.255 any
 exit
interface g0/1.30
 ip access-group GUEST_RESTRICTIONS in
 exit
```

ACL line	What it does
permit udp ... bootpc ... bootps	Lets guests get an IP by DHCP (UDP 68 = DHCP client, UDP 67 = DHCP server)
deny ... 10 / 20 / 99	Blocks guests from Management, POS and Network-Management
permit ip 192.168.30.0 ... any	Allows everything else = Internet only

Step 11: Save the configuration

Do this on both the router and the switch.

```
end
copy running-config startup-config
```

6. End devices

Device	VLAN	IP address	Gateway	DNS
Office Printer	10	192.168.10.10 (static)	192.168.10.1	8.8.8.8
Receipt Printer	20	192.168.20.10 (static)	192.168.20.1	8.8.8.8
Manager PC	10	DHCP (automatic)	192.168.10.1	8.8.8.8
POS Terminal	20	DHCP (automatic)	192.168.20.1	8.8.8.8

Access Point: open its configuration and set an SSID and a pass-phrase (Wi-Fi password). Guest devices then get 192.168.30.x addresses by DHCP.

> **Note:** Printers get static IPs so employees always find them at the same address, which also makes administration and troubleshooting easier. Computers use DHCP.

7. Quick verification

Check	Command / test
VLANs and port assignment	show vlan brief
Trunk up, VLANs allowed	show interfaces trunk
Router sub-interfaces up	show ip interface brief
DHCP pools and leases	show ip dhcp pool | show ip dhcp binding
Guest ACL applied	show access-lists | show ip interface g0/1.30

Check	Command / test
SSH ready (on switch)	show ip ssh
Manager PC works	Ping 192.168.10.1 (gateway) and the office printer 192.168.10.10
Guest isolation works	From a guest device: ping 192.168.10.1 / 192.168.20.1 should FAIL; Internet should work

8. Troubleshooting: new access switch added

An IT company installed a new access switch for the cafe and moved the Access Point to it.

Link	Details
New switch - Multilayer switch	Fast-Ethernet F0/11 on both sides, configured as a trunk
Access Point - New switch	Gigabit port G0/1
Floor jack 1 - New switch	F0/1 (access port, VLAN 30)
Floor jack 2 - New switch	F0/2 (access port, VLAN 30)

Note: On the multilayer switch, F0/11 used to be the Access Point port (VLAN 30 access). It must now be a trunk instead.

Step 1: Identify the problems

Run these on the new switch.

```
show interfaces status      ! interface status
show interfaces trunk       ! trunk link status
show vlan brief             ! access interfaces / VLANs
```

Problem found	Why it matters
Native VLAN mismatch reported on one of the customer-facing interfaces	Compare 'show interfaces trunk' on both ends and make the native VLAN the same on both sides
Small unmanaged switch plugged into floor jack 1 and floor jack 2, with 3 laptops on it	Creates Layer 2 loops and other disasters

Step 2: Enable BPDU Guard on the customer-facing ports

Protects the network from an outside switch being plugged in.

```
enable
configure terminal
interface range f0/1 - 2
 spanning-tree portfast
 spanning-tree bpduguard enable
 exit
```

Note: If these edge ports ever receive a BPDU, the switch disables the port (err-disabled). Fix: remove the unauthorized switch, then recover the port manually with 'shutdown' followed by 'no shutdown'.

Tip: Also check (typical): VLAN 30 exists on the new switch and is allowed on the trunk on both ends; the AP port on the new switch is an access port in VLAN 30.

Appendix: corrections made to the original notes

In the notes	Used in this guide
Router interface 'G0/o'	G0/0
Switch default gateway '1921.68.99.1'	192.168.99.1
'POT terminal'	POS terminal
Receipt Printer 192.168.10.10 with gateway 192.168.20.1 (same IP as the office printer, mixed VLANs)	192.168.20.10 with gateway 192.168.20.1 (POS VLAN 20)
'ip dhcp excluded-addresses'	ip dhcp excluded-address
'inter g0/1.30'	interface g0/1.30
'spanning-tree bpduguard'	spanning-tree bpduguard enable
'power over Internet'	Power over Ethernet (PoE)
Router steps skipped number 9	Steps renumbered 1-11

Added so the commands work (not written in the notes): 'encapsulation dot1Q' on sub-interfaces, 'no shutdown', trunk encapsulation, the SSH command set, and the ISP-side address assumption.